Data and Code for "Regulating Conglomerates: Evidence from an Energy Conservation Program in China"

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Contents

1 Overview **2**

2 Data Availability and Provenance Statement **2**

3 Computational Requirements **6**

4 Description of Programs/Code **7**

5 Instructions for Replicators **7**

1 Overview

- **Data Sources:** This replication package primarily uses data from four sources: the National Bureau of Statistics (NBS, 2001-2009, 2011-2013), Chinese Ministry of Environmental Protection (CMEP, 2001-2010), State Administration of Taxation (SAT, 2007-2011), and State Administration for Industry and Commerce (SAIC, 1949-2018).
- **Software Requirements:** The code in this package relies on Stata (version MP 17) and Matlab (version R2024a).
- **Master Files Included:** There are two master files:
  1. A Stata script, `Do/Master.do`, which generates all exhibits for the main paper and the online appendix.
  2. A Matlab script, `master.m`, which outputs Tables 4, 5, and A36 based on structural analyses using estimated data moments. This program can be run separately.
- **Execution Flow:** The Stata master script, `Do/Master.do`, is designed to call the Matlab master script `master.m`. Running `Do/Master.do` will produce all tables and figures in a streamlined, automated process.
- **Expected Run Time:** The entire replication package is expected to complete within approximately 30-40 minutes.
- **Citation Requirement:** Please cite this replication package when using the data and/or code in your research.

2 Data Availability and Provenance Statement

Statement about Rights: We certify that the authors of the manuscript have legitimate access to and permission to use the data used in this manuscript.

Introduction to Datasets:

This study employs confidential firm-level datasets essential for empirical analysis:

- **Annual Survey of Industrial Firms (ASIF):** Sourced from the National Bureau of Statistics, this dataset provides production and financial information for all industrial firms above designated size in China.
- **China's Environmental Statistics Database (CESD):** Sourced from the Chinese Ministry of Environmental Protection, this dataset provides pollution and energy consumption information for polluting firms in China .
- **Chinese Administration Registration Database (CARD):** Sourced from the State Administration for Industry and Commerce, this dataset provides registration and shareholding information for all firms in China.
- **Enterprise Tax Survey Dataset (ATS):** Sourced from the Chinese State Administration of Tax, this dataset provides tax and financial information for all surveyed firms.

- **Census:** Sourced from the National Bureau of Statistics, this dataset provides production and financial information all industrial firms in China.
- **Patent:** Sourced from the Incopat (https://www.incopat.com/), this dataset provides patent records for all firms in China.
- **Top1,000 and Top10,000 Firms:** Information for the Top1,000 and Top10,000 firms is sourced from the National Development and Reform Commission (NDRC).

In addition to firm-level data, this study also uses macro-level information from several sources:

- **China Statistical Yearbooks:** This includes economic and energy-related information at the provincial, city, and industry level, sourced from the **China Energy Statistical Yearbook (CESY)**, the **China City Statistical Yearbook (CCSY)**, and the **China Industrial Statistical Yearbook (CISY)**.
- **CSMAR Database**. (https://data.csmar.com/) This dataset provides pollution information at the provincial- and city-level in China.

Data Access and Contact Information:

- **Fudan University Data Center Access:** All confidential and yearbook data used in this study are available through Fudan University's Data Center. Researchers seeking replication access may contact:
  - **Contact Person:** Yiyi Xin, Director of the Data Center
  - **Email:** `econdata@fudan.edu.cn`
- **IntSig Information Access:** For firm network data from the Chinese Administrative Registration Database (SAIC), access is available via InSig Information. For inquiries, please contact:
  - **Contact Person:** Lili Hao
  - **Email:** `lili_hao@intsig.net`
- **Replication Support:** The authors support replication requests for up to five years post-publication. For replication assistance, please contact:
  - **Contact Person:** Qiaoyi Chen
  - **Email:** `joychen@fudan.edu.cn`

Details of Raw Data:

This project uses the following data sources:

Table 1: Raw Data Summary

Data file	Source	Notes
Raw_Data/ASIF/ASIF_raw.dta	NBS(2001-2009,2011-2013)	This data includes production and financial information for all industrial firms from the NBS(1998-2009, 2011-2013) files.
Raw_Data/CESD/CESD_raw.dta	CMEP(1998-2010)	This data includes pollution and energy consumption information for all polluting firms from the CMEP(1998-2010) files.
Raw_Data/Census/Census_2004.dta	NBS(2004)	This data includes production and financial information for all industrial firms from the NBS(2004) file.
ATS.dta (Partly provided, see Table 2 for details)	SAT(2007-2011)	This data includes tax and financial information for all firms from the SAT(2007-2011) files.
CARD.dta (Partly provided, see Table 2 for details)	SAIC(1949-2018)	This data includes registration information for all firms from SAIC(1949-2018) files.
Patent.dta (Partly provided, see Table 2 for details)	Incopat(2001-2010)	This data includes patent information for all firms from the Incopat(2001-2010) files.
Raw_Data/Namelist/Top1000_Top10000_ID1(_ID2).dta	NDRC(2006,2011)	This data includes legal code, name and identifiers for all Top1,000 and Top10,000 firms from the NDRC(2006,2011) files.
Raw_Data/Top1000/Top1000_Energy_Saving_Target.dta	NDRC(2008-2010)	This data includes energy saving targets for all Top1,000 firms from the NDRC(2008-2010) files.
Raw_Data/Top1000/Top1000_Industry.dta	NDRC,NBS(2007)	This data includes energy consumption for all Top1,000 firms reported in the bulletin on Top 1,000 firms (2007).
Raw_Data/Top1000/Top1000_Compliance(_Noncompliance).dta	NDRC(2006-2010)	This data includes compliance information for all Top1,000 firms from the NDRC(2006-2010) files.
Raw_Data/Top1000/Top1000_QXB_ID.dta	IntSig(2018)	This data includes basic information for Top1,000 firms from the IntSig Information.
Raw_Data/CARD/Related_Legcode.dta	IntSig(2018)	This data includes basic information for related firms of Top1,000 firms from the IntSig Information.
Raw_Data/Top10000/Top10000_QXB_ID.dta	IntSig(2018)	This data includes basic information for Top10,000 firms from the IntSig Information.
Raw_Data/Top10000/Top10000_(Energy_Saving)_Target(_Legcode/_Name).dta	NDRC(2011)	This data includes energy saving targets for all Top10,000 firms from the NDRC(2011) file.
Raw_Data/Key_Monitor/gas2007_legcode(name).dta	CMEP (2007)	This data includes legal code and name for all monitored air-polluting firms from the CMEP (2007) file.
Raw_Data/Key_Monitor/water2007_legcode(name).dta	CMEP (2007)	This data includes legal code and name for all monitored water-polluting firms from the CMEP (2007) file.
Raw_Data/Industry/Industry_Energy_2001_2011.dta	CESY(2001-2011)	This data includes energy consumption for all 2-digit manufacturing industries from the CESY(2001-2011) files.
Raw_Data/Industry/Industry_Energy_2004.dta	CESY(2004)	This data includes energy consumption for all 2-digit manufacturing industries from the CESY(2004) file.
Raw_Data/Industry/Industry_Energy_Use_2010.dta	CESY(2010)	This data includes energy consumption for all 2-digit manufacturing industries from the CESY(2010) file.
Raw_Data/Industry/Industry_Output_2001_2011.dta	CISY(2001-2011)	This data includes output for all 2-digit manufacturing industries from the CISY(2001-2011) files.
Raw_Data/Industry/Industry_Control_2009_2013.dta	CESY,CISY(2009-2013)	This data includes information for all 2-digit manufacturing industries from the CESY(2009-2013) and CISY(2009-2013) files.
Raw_Data/Industry/Nine_Industry_Energy_Mix.dta	CESY(2001-2010)	This data includes energy mix information for nine energy-intensive industries from the CESY (2001-2010) files.
Raw_Data/Density/City(Province)_Density.dta	CSMAR(2006)	This data includes air pollution density for all cities(provinces) from the CSMAR(2006) file.
Raw_Data/Density/City_Population.dta	CCSY(2006)	This data includes population of all cities from the CCSY(2006) file.

Table 2: Subsamples and Processed Data of Proprietary Raw Data

Data file	Source	Notes
Raw_Data/ATS/ATS_Supplement(_Coal/_Electricity/_Energy/_Other)_Legcode(Name).dta	SAT(2007-2011)	This data includes financial and energy consumption information for all firms from the SAT(2007-2011) files. Processed data from ATS.dta.
Raw_Data/ATS/ATS_Top1000_Top10000.dta	NBS(2001-2008) SAT(2007-2011)	This data includes investment information for Top1,000 and Top10,000 firms from NBS(2001-2008) and SAT(2007-2011) files. Firms that didn't report original value of fixed assets in 2008 or reported different values (by over 25%) in two datasets are excluded. Processed data from Raw_Data/ASIF/ASIF_raw.dta and ATS.dta.
Raw_Data/CESD/Energy_Type.dta	CMEP(1998-2010) SAT(2007-2011)	This data includes main energy types used by each manufacturing industry from the CMEP(1998-2010) files. Processed data from Raw_Data/CESD/CESD_raw.dta and ATS.dta.
Raw_Data/CESD/Industry_Electricity_Share.dta	CMEP(1998-2010) SAT(2007-2011)	This data includes electricity reliance of each manufacturing industry from the CMEP(1998-2010) files. Processed data from Raw_Data/CESD/CESD_raw.dta and ATS.dta.
Raw_Data/Patent/Patent_Top1000_Top10000.dta	Incopat(2001-2010)	This data includes patent information for Top1,000 and Top10,000 firms from the Incopat(2001-2010) files. Processed data from Patent.dta.
Raw_Data/Top1000/Top1000_Capital.dta	SAIC(1949-2018)	This data includes capital information for all Top1,000 firms from the SAIC(1949-2018) files. Processed data from CARD.dta. Provided by the IntSig Information.
Raw_Data/CARD/Related_Capital.dta	SAIC(1949-2018)	This data includes capital information for all related firms of Top1,000 firms from SAIC(1949-2018) files. Processed data from CARD.dta. Provided by the IntSig Information.
Raw_Data/CARD/Top10000_Related_Capital.dta	SAIC(1949-2018)	This data includes capital information for all related firms of Top10,000 firms from the SAIC(1949-2018) files. Processed data from CARD.dta. Provided by the IntSig Information.
Raw_Data/CARD/related_firms(_legcode/_name)_(1-3).dta	SAIC(1949-2018)	This data includes all related manufacturing firms above designated size of Top1,000 firms from SAIC(1949-2018) files. Related firms by up to two(six) levels of investment relations and at least 25%(20%,50%) ownership requirement. Processed data from CARD.dta. Provided by the IntSig Information.
Raw_Data/CARD/Related_Robust.dta	SAIC(1949-2018)	This data includes all related firms of Top1,000 firms from SAIC(1949-2018) files. Processed data from CARD.dta. Provided by the IntSig Information.
Raw_Data/CARD/Related_Firms.dta	SAIC(1949-2018)	This data includes all related pairs of Top1,000 firms and their related firms from SAIC(1949-2018) files. Related firms by up to two levels of investment relations and at least 25% ownership requirement. Processed data from CARD.dta. Provided by the IntSig Information.
Raw_Data/CARD/Conglomerate_ID_Top1000_Legcode(_Name/_Name2).dta	SAIC(1949-2018)	This data includes conglomerate information for Top1,000 firms from SAIC(1949-2018) files. Processed data from CARD.dta. Provided by the IntSig Information.
Raw_Data/CARD/Conglomerate_ID_Related_Legcode(_Name).dta	SAIC(1949-2018)	This data includes conglomerate information for related firms of Top1,000 firms from SAIC(1949-2018) files. Processed data from CARD.dta. Provided by the IntSig Information.
Raw_Data/CARD/Shareholding_Change.dta	SAIC(1949-2018)	This data includes list of all related firms of Top1,000 firms that experienced significant ownership changes between 2007 and 2018 from the SAIC(1949-2018) files. Processed data from CARD.dta. Provided by the IntSig Information.
Raw_Data/Top10000/fudan_affiliates.xlsx	SAIC(1949-2018)	This data includes ID for affiliates of Top10,000 firms from the SAIC(1949-2018) files. Processed data from CARD.dta. Provided by the IntSig Information.
Raw_Data/Top10000/fudan_investment_new.xlsx	SAIC(1949-2018)	This data includes ID for investment of Top10,000 firms from the SAIC(1949-2018) files. Processed data from CARD.dta. Provided by the IntSig Information.
Raw_Data/Top10000/fudan_shareholder_new.xlsx	SAIC(1949-2018)	This data includes ID for shareholders of Top10,000 firms from the SAIC(1949-2018) files. Processed data from CARD.dta. Provided by the IntSig Information.
Raw_Data/Top10000/fudan_shareholder_all.xlsx	SAIC(1949-2018)	This data includes name for shareholders of Top10,000 firms from the SAIC(1949-2018) files. Processed data from CARD.dta. Provided by the IntSig Information.
Raw_Data/Top10000/fudan_shareholder_investment_all_new.xlsx	SAIC(1949-2018)	This data includes ID for firm shareholder investments of Top10,000 firms from the SAIC(1949-2018) files. Processed data from CARD.dta. Provided by the IntSig Information.
Raw_Data/Top10000/fudan_personshareholder_investment_new.xlsx	SAIC(1949-2018)	This data includes ID for peronal shareholder investments of Top10,000 firms from the SAIC(1949-2018) files. Processed data from CARD.dta. Provided by the IntSig Information.
Raw_Data/Top10000/Government_Shareholder.dta	SAIC(1949-2018)	This data includes list of government shareholders for Top10,000 firms from the SAIC(1949-2018) files. Processed data from CARD.dta. Provided by the IntSig Information.

3 Computational Requirements

Software and Hardware Specifications: All analyses were conducted using Stata MP Version 17 and Matlab Version R2024a. The code was last executed on a Lenovo Legion Y7000 2020, equipped with an Intel(R) Core(TM) i7-10750H CPU @ 2.60 GHz and 64 GB of memory. The estimated runtime for all analyses was approximately 30 minutes. The log files from this execution are available in `Log_Stata.smcl`, recorded on October 29, 2024.

Matlab Toolboxes Required: The following Matlab toolboxes are necessary to successfully run the program:

- `Statistical and Machine Learning Toolbox`
- `Optimization Toolbox`
- `Compecon` (Miranda and Fackler, 2011)

Stata Packages Required: The following Stata packages are also required for successful execution. Lines 3-14 in `Master.do` will automatically install these packages if not already available, a backup of each package is stored in `StataAdoBackup/` for convenience.

- `addplot`
- `ebalance`
- `listtex`
- `unique`
- `binscatter`
- `ftools`
- `estout`
- `winsor2`
- `reghdfe`
- `fdta`
- `parmest`
- `xfill`
- `texdoc`

4 Description of Programs/Code

Code Structure: Code is organized into four main directories:

- `Data_Generation_Program`: Contains routines for data cleaning and the creation of analysis files. Outputs are saved in `Data`. Note: This code relies on proprietary data.
- `Do`: Contains routines to generate tables, graphs, and scalar values. These outputs are saved in `Table`, `Figure`, `Appendix_Table` and `Appendix_Figure`. Code in this folder relies on data stored in the `Data` folder.
- `Data_Moments/Do`: Contains routines to produce data moments required for structural estimation. Outputs are saved in `Data_Moments/Result`. This code relies on data in the `Data` folder.
- `Structural_Analysis`: Contains routines for structural analyses. Outputs are saved in `Structural_Analysis/Output`. Code in this folder relies on estimated data moments generated in `Data_Moments/Result`.

Master File: One master file runs all analyses for the paper:

- Run `Do/Master.do` file to complete all the following steps:
    1. **Data Cleaning in Stata**: Calls scripts listed in Table 3 and complete data cleaning and analysis file generation routines. This section requires proprietary data, which can be found in `Raw_Data` for authorized users.
    2. **Structural Estimation in Matlab**: Calls scripts listed in Table 4 and complete the structural estimation routines.
    3. **Graphs and Tables in Stata**: Call scripts listed in Table 6 and generates graphs and tables. This section uses analysis data files located in `Data`.

Outputs: Results are deposited in the following directories:

- **Reduced-Form Outputs:** Saved in `Figure`, `Table`, `Appendix_Figure`, and `Appendix_Table`.
- **Structural Estimation Outputs:** Saved in `Structural_Analysis/Output`.

5 Instructions for Replicators

To replicate the results of this paper, please follow the steps below:

- **Set Working Directories:** Use the following Stata script to run the primary master file: `Do/Master.do`. Before running this script, replicators need to set the working directories at the specified locations in both Stata and Matlab code:
    1. Modify Line 17 of `Do/Master.do` to set the path for the Stata scripts.

2. Update Lines 4 and 5 in `Structural_Analysis/define_directory.m` to set paths for the Matlab scripts. Note the path structure in these lines: the paths should follow the format `PATH/Structural_Analysis`, where `PATH` represents your specific directory path.

- **Trial Run:** Since the Matlab master script, `Structural_Analysis/master.m`, is executed from within `Do/Master.do`, it is essential to specify the Matlab executable on Line 29 of `Do/Master.do`. Without this, `master.m` and additional scripts (`Table2.do`, `TableA24.do`, and `TableA38.do`) will not execute correctly.

  1. In most cases, setting Line 27 as follows will work: `global Matlab "!Matlab"`.
  2. If you encounter an error such as `zsh command not found: Matlab`, this indicates that UNIX/Terminal cannot locate the Matlab executable. In this case, you must specify the full path to the Matlab executable. For example, MacOS users with Matlab R_2023b should set Line 27 as follows:
     `global Matlab ‘‘!/Applications/Matlab_R2023b.app/bin/Matlab’’`

- **Path Formatting:** When setting working directories, ensure correct path formatting, noting when a directory ends with a forward slash ("/") and when it does not. Incorrect path formatting may prevent the code from running properly.

Known Issues

- `Table2.do`, `TableA24.do`, and `TableA38.do` contain lines that call Matlab scripts within the Stata do-file. For instance:
  - Line 100 of `Table2.do` runs `Distance_Matching_Output.m`
  - Line 322 of `Table2.do` runs `Distance_Matching_Placebo.m`
  - Line 549 of `Table2.do` runs `Distance_Matching_Energy.m`

- If Matlab can be called with Line 29 of `Do/Master.do` and displays "STATA Configuration Completed!", the remaining code will execute successfully.

Table 3: Cleaning Code

Code	Purpose of Code
Data_Generation_Program/Data_Cleaning.do	Add Top1,000 and Top10,000 firm identifiers to ASIF and CESD, merge CESD with ASIF, and conduct basic cleaning routines, saves in Raw_Data.
Data_Generation_Program/Data_for_Table1.do	Prepares Data for Table Table 1, A5, A7 Panel A, A10, A11, A12, A13, A15, A16, A17, A18, A19, A20, A21, A32, A34 and Figure 3, A6, A9, A10, A14, A19, saves it in Data
Data_Generation_Program/Data_for_Table2.do	Prepares Data for Table 2, A25, A27, A28, A29, A35 and Figure 4, A16, A18, A20, saves it in Data
Data_Generation_Program/Data_for_Table3.do	Prepares Data for Table 3, A30 and Figure 2 Panel B, 5, saves it in Data
Data_Generation_Program/Data_for_TableA1.do	Prepares Data for Table A1, saves it in Data
Data_Generation_Program/Data_for_TableA2.do	Prepares Data for Table A2, saves it in Data
Data_Generation_Program/Data_for_TableA3.do	Prepares Data for Table A3, saves it in Data
Data_Generation_Program/Data_for_TableA4.do	Prepares Data for Table A4, saves it in Data
Data_Generation_Program/Data_for_TableA6.do	Prepares Data for Table A6, saves it in Data
Data_Generation_Program/Data_for_TableA7.do	Prepares Data for Table A7 Panel B, saves it in Data
Data_Generation_Program/Data_for_TableA8.do	Prepares Data for Table A8, A9 Panel A and Figure A8, saves it in Data
Data_Generation_Program/Data_for_TableA9.do	Prepares Data for Table A9 Panel B, saves it in Data
Data_Generation_Program/Data_for_TableA14.do	Prepares Data for Table A14 and Figure A11, saves it in Data
Data_Generation_Program/Data_for_TableA22.do	Prepares Data for Table A22 and Figure A15, saves it in Data
Data_Generation_Program/Data_for_TableA23.do	Prepares Data for Table A23, saves it in Data
Data_Generation_Program/Data_for_TableA24.do	Prepares Data for Table A24, saves it in Data
Data_Generation_Program/Data_for_TableA26.do	Prepares Data for Table A26 and Figure A17, saves it in Data
Data_Generation_Program/Data_for_TableA37.do	Prepares Data for Table A37 and Figure A22, saves it in Data
Data_Generation_Program/Data_for_TableA38.do	Prepares Data for Table A38, saves it in Data
Data_Generation_Program/Data_for_FigureA5.do	Prepares Data for Figure A5, saves it in Data
Data_Generation_Program/Data_for_FigureA12.do	Prepares Data for Figure A12, saves it in Data
Data_Generation_Program/Data_for_FigureA13.do	Prepares Data for Figure A13, saves it in Data
Data_Generation_Program/Data_for_Moment1.do	Prepares Data for Moment 1, saves it in Data
Data_Generation_Program/Data_for_Moment3.do	Prepares Data for Moment 3 and Figure 2 Panel A, saves it in Data
Data_Generation_Program/Data_for_Scalars.do	Prepares Data for Scalars, saves it in Data

Table 4: Structural Estimation Code

Type of Analysis	Code Use
1. Structural Moments	Outputs are saved in `Data_Moments/Result`
This file generates moments for firm size distribution.	`Data_Moments/Do/Moment1.do`
This file generates moments for within-conglomerate distribution.	`Data_Moments/Do/Moment2.do`
This file generates moments for share of small firms.	`Data_Moments/Do/Moment3.do`
2. Structural Estimation	`Structural_Analysis/master.m`
This file runs the following structural analysis:	
2.1 Baseline Estimation	Located
Estimate parameters, output Table 4	`Structural_Analysis/Estimation/Main.m`
Compare model fits, output Figure 6	
2.2 Estimation: Robustness	Located
Alternative combinations ρ and σ	`Structural_Analysis/Estimation_Menu/Main.m`
Imperfect substitution	`Structural_Analysis/Estimation_ImpSub/Main.m`
	`Structural_Analysis/Estimation_ImpSub/Main_ste.m`
2.3 Sensitivity Analysis	Located
Outputs TableA36	`Structural_Analysis/Sensitivity Analysis/Main_ste.m`
	`Structural_Analysis/Sensitivity Analysis/Main_ste_menu.m`
2.4 Counterfactuals	Located
Output Table 5	`Structural_Analysis/Counterfactual Baseline/Main.m`
Output Figure 7	`Structural_Analysis/Counterfactual Baseline/Main_compare.m`
Output Figure A2 Panel C	`Structural_Analysis/Counterfactual Robustness/Main_compare.m`
Output Figure A3	
2.5 Robustness: Energy Efficiency	Located
Output Figure A2 Panel A	`Structural_Analysis/Energy Efficiency/Main.m`
2.6 Robustness: Heterogeneous Energy Efficiency	Located
Output Figure A2 Panel B	`Structural_Analysis/Heterogeneous EE/Main.m`
Output Figure A21	`Structural_Analysis/Heterogeneous EE/Main_Robust.m`
2.7 Robustness: Imperfect Substitution	Located
Output Figure A2 Panel C	`Structural_Analysis/Imperfect Sub/Main.m`
2.8 Robustness: Capital Adjustment	Located
Output Figure A2 Panel C	`Structural_Analysis/Counterfacutal AdjK/Main.m`
2.9 Robustness: No Conglomerate	Located
Output Figure A2 Panel C	`Structural_Analysis/No Conglomerate/Main.m`

Table 5: Result of Structural Estimation that is taken as input by Stata

Estimation Result File in `Structural_Analysis/`	Stata `.do` script that uses the result	Output Exhibit
`Estimation/moments.txt`	`Do/Figure6.do`	Figure 6 Panel A
`Counterfactual Baseline/results_figure_6B.txt`	`Do/Figure6.do`	Figure 6 Panel B
`Counterfactual Baseline/results.txt`	`Do/Figure7.do`	Figure 7 Panel A, B
`Energy Efficiency/results.txt`	`Do/FigureA2.do`	Figure A2 Panel A
`Heterogeneous EE/results.txt`	`Do/FigureA2.do`	Figure A2 Panel B
`Imperfect Sub/results.txt`	`Do/FigureA2.do`	Figure A2 Panel C
`Counterfactual Robustness/results_85_75.txt`	`Do/FigureA2.do`	Figure A2 Panel C
`Counterfactual Robustness/results_90_70.txt`	`Do/FigureA2.do`	Figure A2 Panel C
`Counterfactual Robustness/results_90_90.txt`	`Do/FigureA2.do`	Figure A2 Panel C
`Counterfactual Robustness/results_95_75.txt`	`Do/FigureA2.do`	Figure A2 Panel C
`Counterfactual AdjK/results.txt`	`Do/FigureA2.do`	Figure A2 Panel C
`No Conglomerate/results.txt`	`Do/FigureA2.do`	Figure A2 Panel C
`Counterfactual Baseline/results.txt`	`Do/FigureA2.do`	Figure A2 Panel C
`Counterfactual Baseline/results_graphs.txt`	`Do/FigureA3.do`	Figure A3 Panel A, B
`Heterogeneous EE/results_robust.txt`	`Do/FigureA21.do`	Figure A21

Table 6: Code for Graphs and Tables[†]

Code	Purpose of Code
Do/Figure1.do	Saves Figure 1 in Figure
Do/Figure2.do	Saves Figure 2 in Figure
Do/Figure3.do	Saves Figure 3 in Figure
Do/Figure4.do	Saves Figure 4 in Figure
Do/Figure5.do	Saves Figure 5 in Figure
Do/Figure6.do	Saves Figure 6 in Figure
Do/Figure7.do	Saves Figure 7 in Figure
Do/Table1.do	Saves Table 1 in Table
Do/Table2.do	Saves Table 2 in Table
Do/Table3.do	Saves Table 3 in Table
Structural_Analysis/Output/Table4.m	Saves Table 4 in Structural_Analysis/Output/Table4.txt
Structural_Analysis/Output/Table5.m	Saves Table 5 in Structural_Analysis/Output/Table5.txt
Do/FigureA2.do	Saves Figure A2 in Appendix_Figure
Do/FigureA3.do	Saves Figure A3 in Appendix_Figure
Do/FigureA5.do	Saves Figure A5 in Appendix_Figure
Do/FigureA6.do	Saves Figure A6 in Appendix_Figure
Do/FigureA8.do	Saves Figure A8 in Appendix_Figure
Do/FigureA9.do	Saves Figure A9 in Appendix_Figure
Do/FigureA10.do	Saves Figure A10 in Appendix_Figure
Do/FigureA11.do	Saves Figure A11 in Appendix_Figure
Do/FigureA12.do	Saves Figure A12 in Appendix_Figure
Do/FigureA13.do	Saves Figure A13 in Appendix_Figure
Do/FigureA14.do	Saves Figure A14 in Appendix_Figure
Do/FigureA15.do	Saves Figure A15 in Appendix_Figure
Do/FigureA16.do	Saves Figure A16 in Appendix_Figure
Do/FigureA17.do	Saves Figure A17 in Appendix_Figure
Do/FigureA18.do	Saves Figure A18 in Appendix_Figure
Do/FigureA19.do	Saves Figure A19 in Appendix_Figure
Do/FigureA20.do	Saves Figure A20 in Appendix_Figure
Do/FigureA21.do	Saves Figure A21 in Appendix_Figure
Do/FigureA22.do	Saves Figure A22 in Appendix_Figure
Do/TableA1.do	Saves Table A1 in Appendix_Table
Do/TableA2.do	Saves Table A2 in Appendix_Table
Do/TableA3.do	Saves Table A3 in Appendix_Table
Do/TableA4.do	Saves Table A4 in Appendix_Table
Do/TableA5.do	Saves Table A5 in Appendix_Table
Do/TableA6.do	Saves Table A6 in Appendix_Table
Do/TableA7.do	Saves Table A7 in Appendix_Table
Do/TableA8.do	Saves Table A8 in Appendix_Table
Do/TableA9.do	Saves Table A9 in Appendix_Table
Do/TableA10.do	Saves Table A10 in Appendix_Table
Do/TableA11.do	Saves Table A11 in Appendix_Table
Do/TableA12.do	Saves Table A12 in Appendix_Table
Do/TableA13.do	Saves Table A13 in Appendix_Table
Do/TableA14.do	Saves Table A14 in Appendix_Table
Do/TableA15.do	Saves Table A15 in Appendix_Table
Do/TableA16.do	Saves Table A16 in Appendix_Table
Do/TableA17.do	Saves Table A17 in Appendix_Table
Do/TableA18.do	Saves Table A18 in Appendix_Table
Do/TableA19.do	Saves Table A19 in Appendix_Table
Do/TableA20.do	Saves Table A20 in Appendix_Table
Do/TableA21.do	Saves Table A21 in Appendix_Table
Do/TableA22.do	Saves Table A22 in Appendix_Table
Do/TableA23.do	Saves Table A23 in Appendix_Table
Do/TableA24.do	Saves Table A24 in Appendix_Table
Do/TableA25.do	Saves Table A25 in Appendix_Table
Do/TableA26.do	Saves Table A26 in Appendix_Table
Do/TableA27.do	Saves Table A27 in Appendix_Table
Do/TableA28.do	Saves Table A28 in Appendix_Table
Do/TableA29.do	Saves Table A29 in Appendix_Table
Do/TableA30.do	Saves Table A30 in Appendix_Table
Do/TableA31.do	Saves Table A31 in Appendix_Table
Do/TableA32.do	Saves Table A32 in Appendix_Table
Do/TableA33.do	Saves Table A33 in Appendix_Table
Do/TableA34.do	Saves Table A34 in Appendix_Table
Do/TableA35.do	Saves Table A35 in Appendix_Table
Structural_Analysis/Output/TableA36.m	Saves Table A36 in Structural_Analysis/Output/TableA36.txt
Do/TableA37.do	Saves Table A37 in Appendix_Table
Do/TableA38.do	Saves Table A38 in Appendix_Table

Notes: [†] Figure A1 and A7 in the paper are examples of related firms which can be plotted directly based on CARD. Figure A4 is a record for interviews conducted by the authors.
Table 4, 5 and A36 are produced by Matlab routines. Figure 6, 7, A2, A3, and A21 are produced based on Matlab results. Table A33 is calculations based on the literature and the world bank report.

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